RESULTS AND DISCUSSION

1 Exploratory Analysis Results

The exploratory analysis of the integrated Chicago crime dataset provided valuable insights into the spatial, temporal, environmental, and socio-economic factors associated with crime incidents and arrest outcomes. By combining crime records with weather observations, public transit mobility, and socio-economic indicators, the exploratory analysis established a comprehensive understanding of the dataset and identified several meaningful patterns that justified the selection of features for the predictive models.

1.1 Descriptive and Temporal Analysis

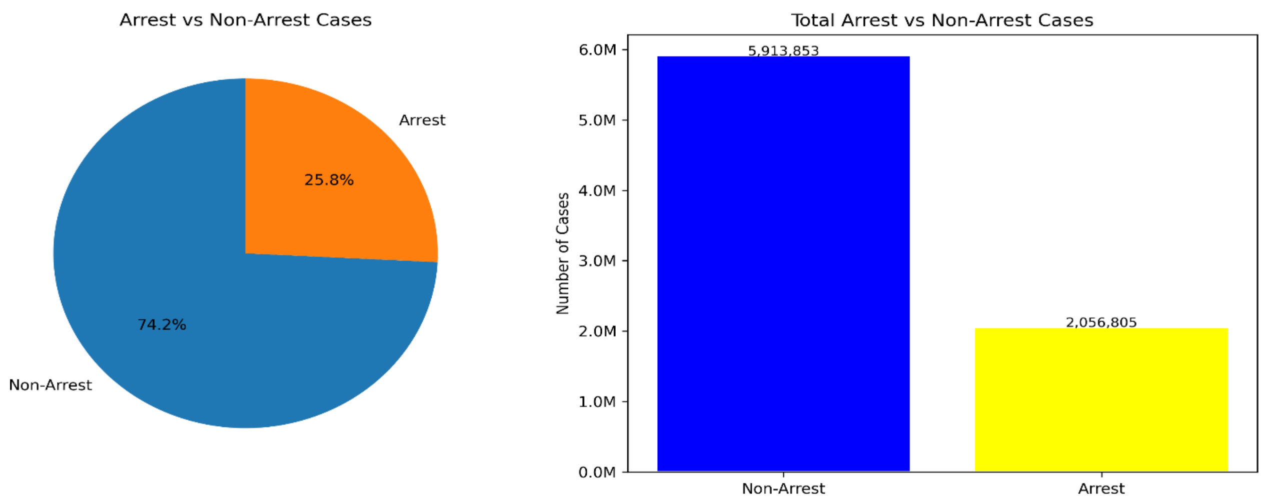

Figure 1: Arrest vs. Non-Arrest Distribution

The analysis first revealed that the dataset was naturally imbalanced, with non-arrest incidents accounting for approximately 74.20% of all crime records, while arrest cases represented only 25.80%. This imbalance indicated that classification models trained directly on the original dataset would likely become biased toward the majority class, highlighting the necessity of investigating appropriate sampling techniques during the machine learning experiments.

Temporal analysis demonstrated that crime occurrence varied considerably across different years, months, days of the week, and hours of the day. A noticeable reduction in reported crime incidents was observed during the COVID-19 pandemic, followed by a gradual increase as public activities resumed. Monthly and weekly analyses showed recurring seasonal trends, while hourly analysis indicated that crime activity was concentrated during specific periods of the day. These observations confirmed that temporal variables such as year, month, day of the week, and hour contain important predictive information and should be incorporated into the arrest prediction framework.

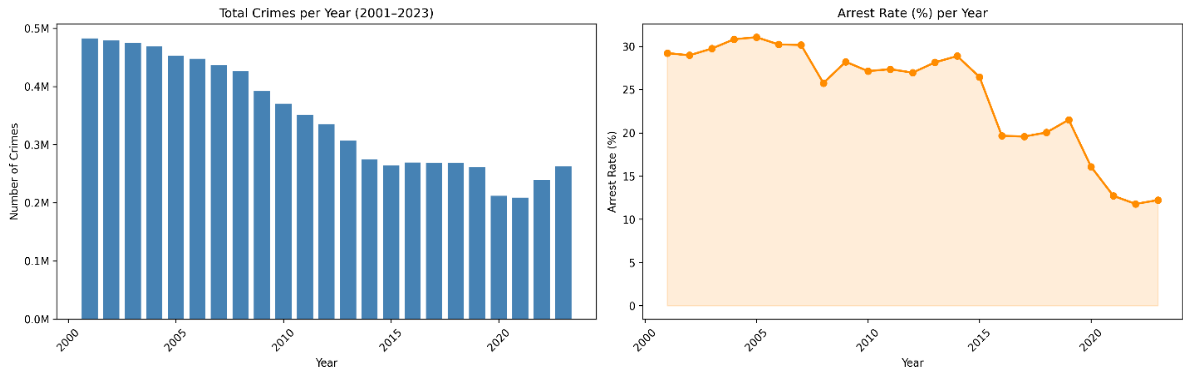

Figure 2: Crimes and Arrest Rate per Year

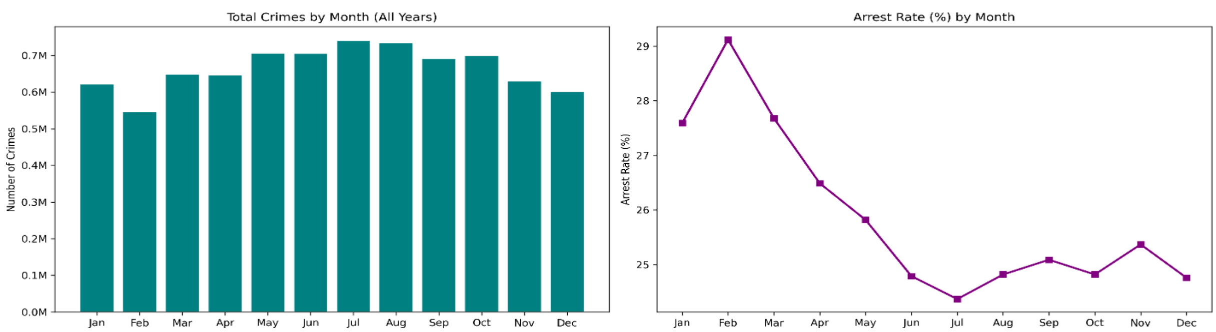

Figure 3: Crimes and Arrest Rate by Month

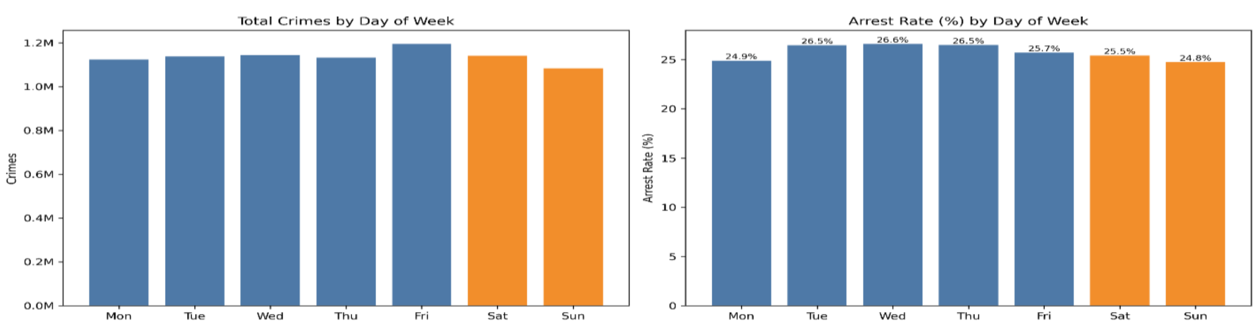

Figure 4: Crimes and Arrest Rate by Day of Week

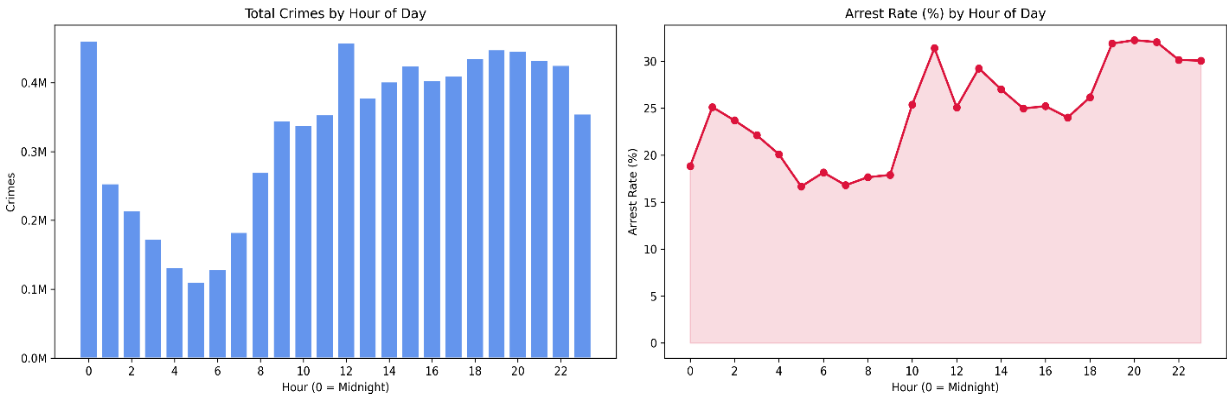

Figure 5: Crimes and Arrest Rate by Hour

1.2 Spatial and Socio-Economic Analysis

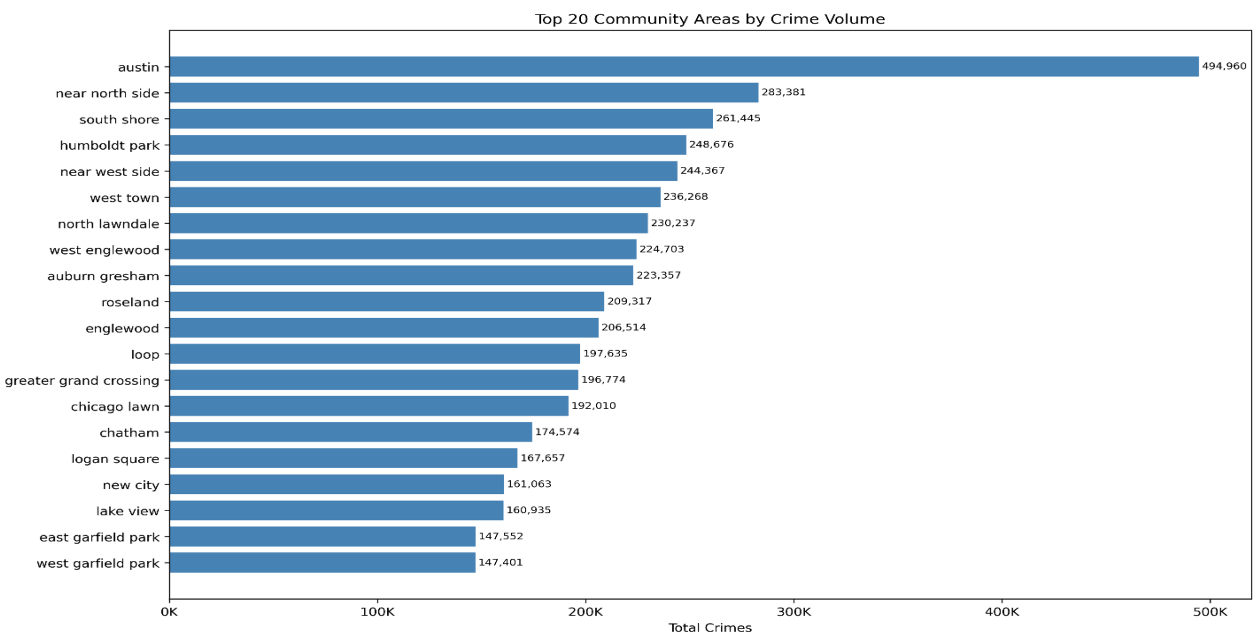

Figure 6: Top 20 Community Areas by Crime Volume

Spatial analysis further demonstrated that crime incidents were not uniformly distributed throughout Chicago. Instead, crimes were concentrated within particular community areas, police districts, and beats, indicating the presence of strong geographic patterns. Such spatial clustering supports previous criminological findings that crime is influenced by local environmental characteristics and justifies the inclusion of geographic variables, including latitude, longitude, police beat, district, and community area, within the predictive models.

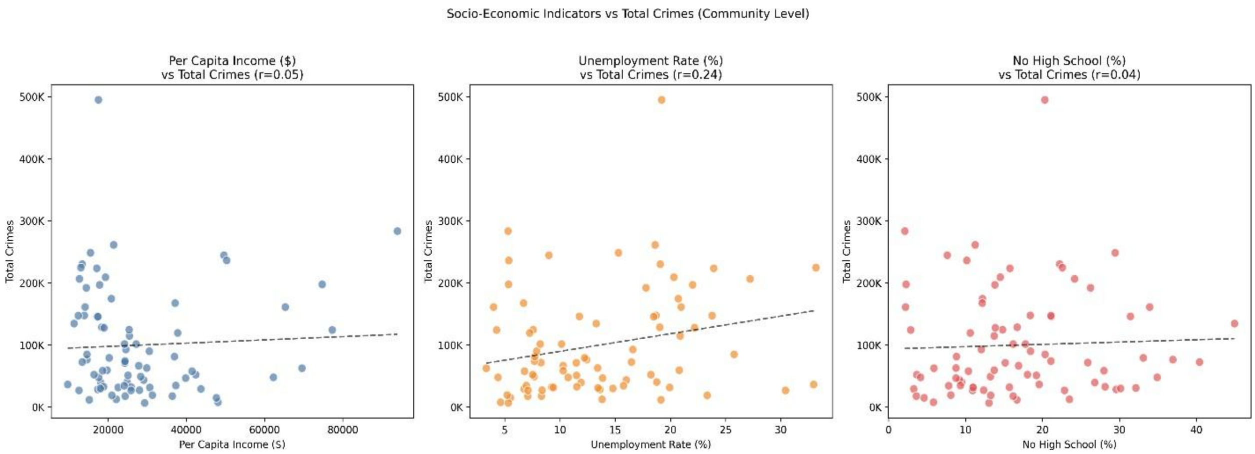

Figure 7: Socio-economic Factors vs. Total Crimes

The integration of socio-economic variables provided additional insight into neighbourhood-level differences associated with crime incidents. Community areas with varying levels of unemployment, educational attainment, poverty, and per capita income exhibited different crime characteristics, suggesting that socio-economic conditions contribute to explaining variations in arrest outcomes. These findings support the inclusion of demographic and economic indicators as contextual variables rather than relying solely on crime-specific information.

1.3 Environmental and Mobility Interaction Analysis

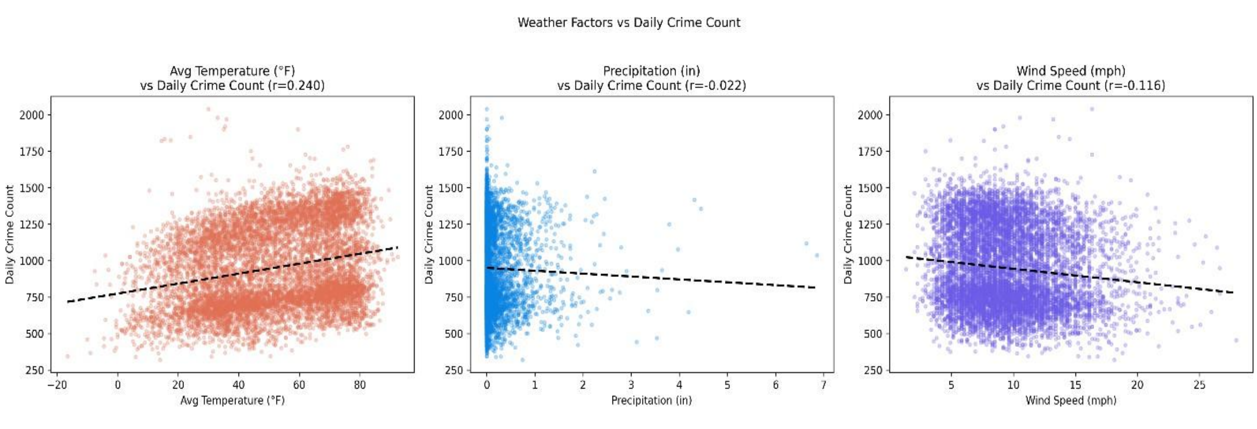

Figure 8: Weather Factors vs. Daily Crime Count

The exploratory analysis also demonstrated that environmental conditions influence crime patterns. Weather variables such as maximum temperature, minimum temperature, rainfall, snowfall, wind speed, and derived temperature categories displayed observable seasonal variations that corresponded with fluctuations in crime frequency. Although weather alone does not determine criminal activity, its interaction with temporal and spatial factors provides additional predictive information that enhances model performance.

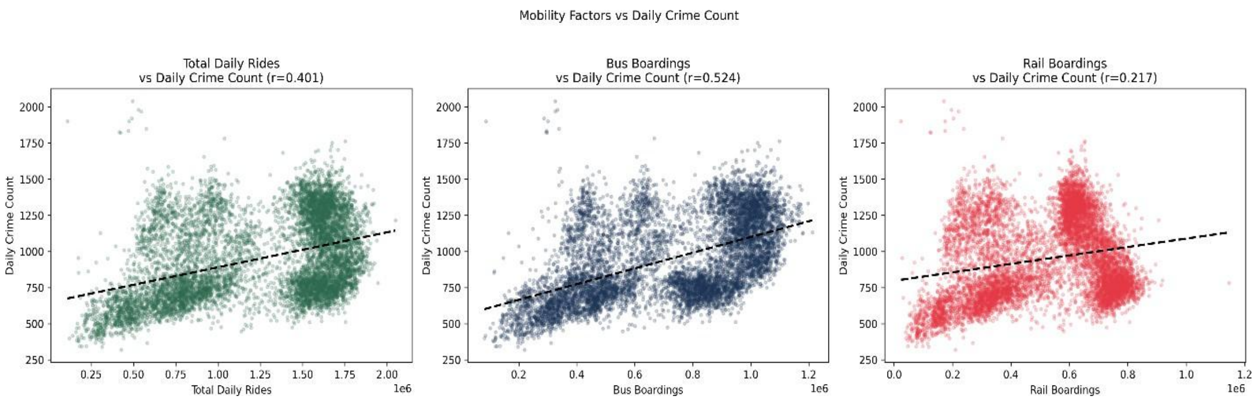

Figure 9: Public Transit Mobility vs. Daily Crime Count

Public transit mobility analysis revealed a moderate positive relationship between daily transit usage and reported crime incidents. Bus ridership exhibited the strongest positive correlation with crime occurrence, followed by total transit rides and rail boardings. Furthermore, comparison of the pre-pandemic, pandemic, and post-pandemic periods showed that significant reductions in public mobility coincided with lower crime counts, while the gradual recovery of transit usage was accompanied by increasing crime activity. These observations suggest that human mobility captures changes in urban activity levels and represents an important complementary predictor for arrest prediction.

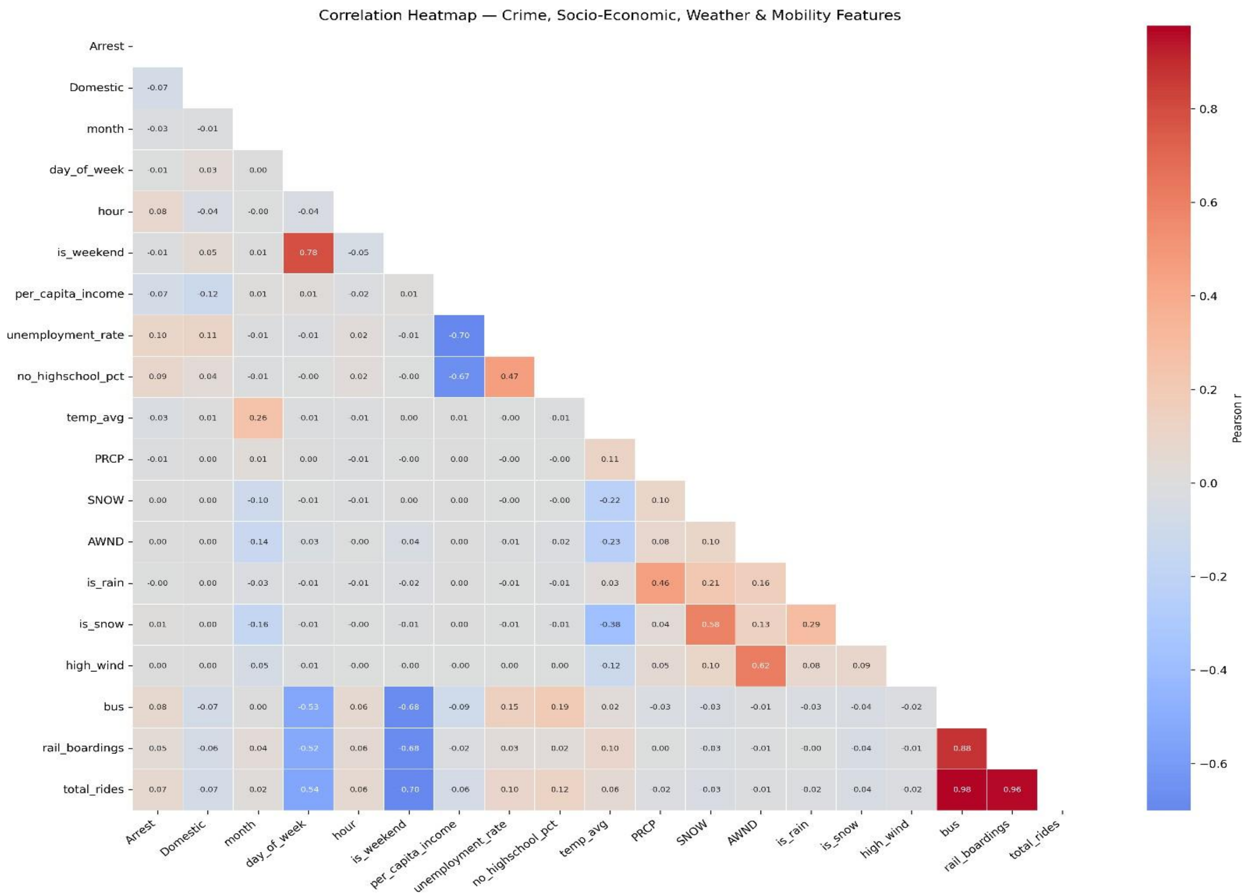

Figure 10: Correlation Heatmap of Selected Features

The correlation analysis showed that while no single variable exhibited a dominant linear relationship with arrest outcomes, several features demonstrated meaningful associations with crime occurrence and urban activity. The combination of temporal, spatial, socio-economic, weather, and mobility variables provided complementary information, reinforcing the importance of integrating multiple data sources rather than relying on a single category of predictors.

Overall, the exploratory analysis confirmed that arrest prediction is influenced by the interaction of multiple factors rather than any single variable. Temporal, spatial, socio-economic, weather, and mobility characteristics each contributed unique information describing crime behaviour across the city. Consequently, these findings provided strong justification for integrating multi-source data into the framework and guided the selection of features used during the machine learning experiments.

2 Machine Learning Experiment Results

A series of machine learning experiments was conducted to develop and evaluate the arrest prediction framework. Since the original Chicago crime dataset exhibited a significant class imbalance, multiple experimental settings were investigated to determine the most appropriate strategy for predicting arrest outcomes. Three experimental scenarios were evaluated using the original imbalanced dataset, a randomly undersampled dataset, and a randomly oversampled dataset. Each experiment employed the same feature set while varying only the sampling strategy to assess its impact on classification performance.

2.1 Experiment 1: Original Imbalanced Dataset

The first experiment established the baseline performance of the framework using the original dataset without applying any class balancing technique.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	PR AUC	Time (s)
Random Forest	0.8936	0.9086	0.6534	0.7602	0.9116	0.8549	372.37
Extra Trees	0.8889	0.8917	0.6483	0.7507	0.9047	0.8449	304.95
LightGBM	0.8900	0.9111	0.6358	0.7490	0.9109	0.8532	19.73
XGBoost	0.8896	0.9068	0.6375	0.7487	0.9112	0.8535	21.11
TCN	0.8872	0.8941	0.6387	0.7451	0.9073	0.8486	3560.00
CatBoost	0.8867	0.9043	0.6272	0.7407	0.9045	0.8446	31.83
LSTM	0.8855	0.8923	0.6326	0.7403	0.9046	0.8442	568.50
GRU	0.8851	0.8900	0.6330	0.7398	0.9038	0.8431	571.07
MLP	0.8833	0.8893	0.6258	0.7347	0.9004	0.8384	739.46
Gradient Boosting	0.8817	0.9000	0.6095	0.7268	0.8976	0.8365	2829.69
AdaBoost	0.8665	0.8884	0.5519	0.6808	0.8768	0.7984	1102.77
Decision Tree	0.8316	0.6668	0.6940	0.6801	0.7868	0.5419	222.75
SGD Linear SVM	0.8220	0.7231	0.5024	0.5929	0.7333	0.5037	77.51
Linear SVM	0.8198	0.7228	0.4896	0.5838	0.7471	0.5261	196.27
Logistic Regression	0.8181	0.7222	0.4798	0.5766	0.7509	0.5388	5.32

The results demonstrated that most machine learning models achieved relatively high overall accuracy because the majority of crime incidents belonged to the non-arrest class. However, despite the high accuracy, several models produced comparatively lower recall values for arrest prediction, indicating difficulty in correctly identifying minority-class observations. These findings confirmed that relying solely on accuracy is insufficient for evaluating classifier performance on imbalanced datasets and highlighted the need for additional techniques to improve minority-class prediction.

2.2 Experiment 2: Random Undersampling

The second experiment applied Random Undersampling to balance the class distribution by reducing the number of majority-class observations before model training.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	PR AUC	Time (s)
TCN	0.8750	0.7950	0.6950	0.7416	0.9033	0.8424	940.99
XGBoost	0.8640	0.7313	0.7479	0.7395	0.9113	0.8527	9.85
Random Forest	0.8604	0.7140	0.7659	0.7391	0.9133	0.8545	162.06
LightGBM	0.8630	0.7284	0.7479	0.7380	0.9109	0.8520	9.35
CatBoost	0.8624	0.7318	0.7368	0.7343	0.9054	0.8440	15.71
LSTM	0.8680	0.7702	0.6960	0.7312	0.8981	0.8335	62.38
Gradient Boosting	0.8628	0.7409	0.7202	0.7304	0.8990	0.8360	1586.42
GRU	0.8692	0.7835	0.6817	0.7290	0.8939	0.8271	57.56
Extra Trees	0.8533	0.6969	0.7637	0.7288	0.9070	0.8453	146.47
MLP	0.8496	0.6920	0.7515	0.7205	0.9013	0.8389	163.12
AdaBoost	0.8326	0.6721	0.6861	0.6790	0.8743	0.7949	589.72
Decision Tree	0.7698	0.5372	0.7798	0.6361	0.7731	0.4758	93.87
Logistic Regression	0.7479	0.5092	0.6403	0.5673	0.7546	0.5247	2.62
Linear SVM	0.7478	0.5090	0.6397	0.5669	0.7539	0.5244	32.04
SGD Linear	0.7417	0.4995	0.6297	0.5571	0.7389	0.4913	36.15

The undersampling strategy substantially improved the ability of most classifiers to recognize arrest cases, resulting in higher recall and F1-score compared with the baseline experiment. Ensemble learning methods, particularly Random Forest, XGBoost, LightGBM, CatBoost, Gradient Boosting, and Extra Trees, consistently produced strong predictive performance across multiple evaluation metrics. The balanced class distribution enabled the classifiers to learn minority-class characteristics more effectively, although reducing the size of the majority class also resulted in some loss of information.

2.3 Experiment 3: Random Oversampling

The third experiment employed Random Oversampling by increasing the number of minority-class observations through duplication of arrest cases while preserving all majority-class records.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	PR AUC	Time (s)
Random Forest	0.8908	0.8723	0.6760	0.7617	0.9122	0.8545	631.86
Extra Trees	0.8890	0.8965	0.6440	0.7496	0.9044	0.8444	555.41
LSTM	0.8709	0.7708	0.7110	0.7397	0.9049	0.8441	401.20
XGBoost	0.8639	0.7305	0.7489	0.7396	0.9115	0.8530	25.44
LightGBM	0.8635	0.7303	0.7469	0.7385	0.9108	0.8520	16.59
TCN	0.8752	0.8063	0.6798	0.7376	0.8999	0.8372	865.74

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	PR AUC	Time (s)
CatBoost	0.8623	0.7317	0.7365	0.7341	0.9057	0.8443	39.30
GRU	0.8681	0.7725	0.6929	0.7305	0.8956	0.8303	108.93
Gradient Boosting	0.8627	0.7429	0.7157	0.7291	0.8986	0.8352	4069.94
MLP	0.8530	0.7026	0.7464	0.7238	0.9026	0.8398	612.00
Decision Tree	0.8341	0.6748	0.6892	0.6819	0.7869	0.5454	429.38
AdaBoost	0.8295	0.6624	0.6919	0.6768	0.8758	0.7973	1584.49
Logistic Regression	0.7478	0.5091	0.6404	0.5672	0.7547	0.5248	10.60
Linear SVM	0.7477	0.5089	0.6399	0.5669	0.7540	0.5245	155.42
SGD Linear SVM	0.7416	0.4994	0.6289	0.5567	0.7387	0.4902	98.49

Compared with the original dataset, oversampling further improved minority-class prediction by allowing classifiers to learn more representative arrest patterns. Several ensemble models maintained high predictive performance, while some algorithms exhibited slight improvements in recall and F1-score. However, the increased training data also resulted in longer computational time and a greater possibility of overfitting because duplicated minority-class observations were repeatedly presented during training.

3 Comparative Performance Analysis

The performance of the evaluated machine learning and deep learning models was compared across the three sampling strategies to identify the most effective approach for arrest prediction. While the previous subsections presented the detailed numerical performance of each experiment, this section provides a comprehensive comparison using graphical visualizations that facilitate interpretation of the differences among models and sampling techniques.

The comparison focuses on Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC, allowing the strengths and limitations of each approach to be examined from multiple perspectives. Since arrest prediction represents an imbalanced binary classification problem, greater emphasis is placed on F1-score, ROC-AUC, and PR-AUC rather than overall Accuracy.

3.1 Average Performance by Sampling Method

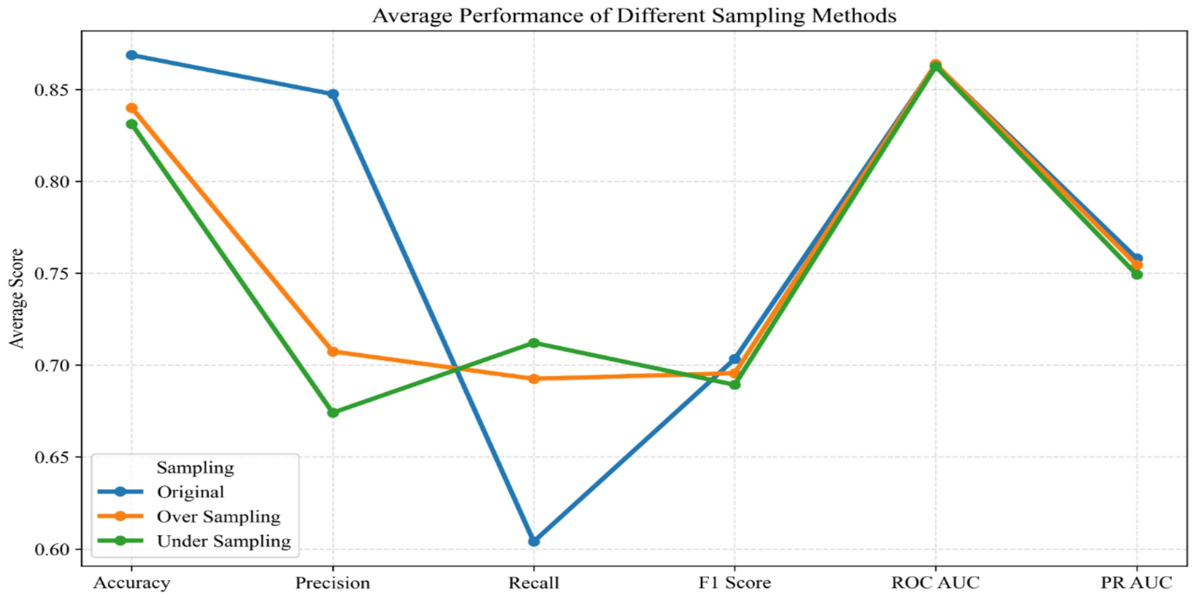

Figure 11: Average Performance of Different Sampling Methods

Figure 11 presents the average classification performance achieved by all evaluated models under the three sampling strategies. The results indicate that the original dataset achieved the highest average Accuracy and Precision because the models were predominantly influenced by the majority non-arrest class. However, this improvement in Accuracy did not necessarily translate into better identification of arrest cases.

Both Random Undersampling and Random Oversampling substantially improved the average Recall and F1-score by providing a more balanced representation of the minority arrest class during training. Among the two balancing approaches, Random Oversampling achieved slightly better overall performance across most evaluation metrics while preserving all observations from the original dataset. In contrast, Random Undersampling reduced the size of the majority class, resulting in faster model training but at the cost of discarding potentially useful information.

Overall, the comparison demonstrates that addressing class imbalance improves the ability of predictive models to identify arrest cases more effectively, while Random Oversampling provides the best balance between minority-class detection and overall classification performance.

3.2 Top Performing Models

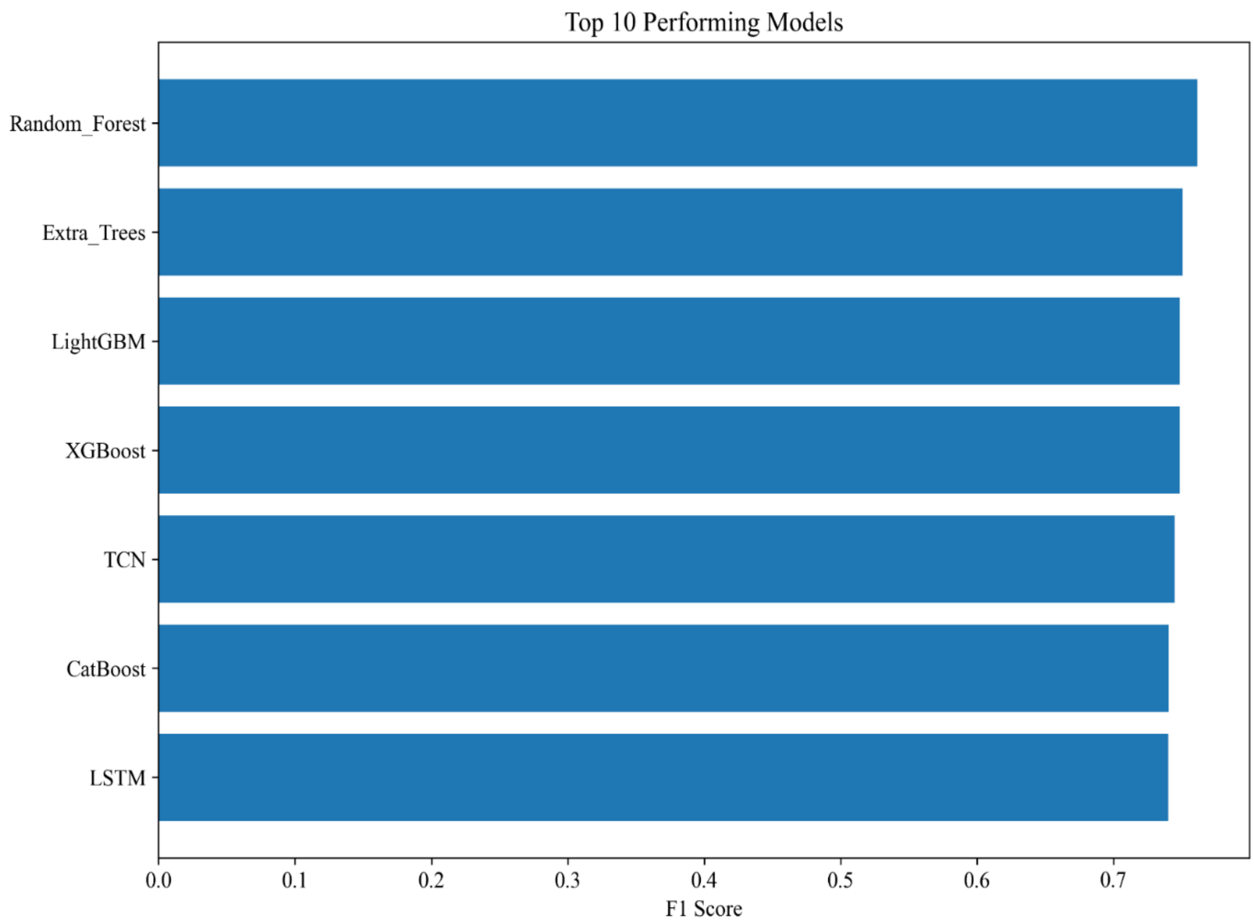

Figure 12: Top 10 Performing Models

Figure 12 compares the ten best-performing models ranked according to their overall predictive performance. Ensemble learning algorithms consistently dominate the ranking, with Random Forest, XGBoost, LightGBM, CatBoost and Extra Trees occupying most of the top positions. These models effectively capture complex non-linear relationships among crime characteristics, temporal information, socio-economic indicators, weather conditions, and public transit mobility.

Random Forest achieved the strongest overall performance by maintaining an excellent balance between Precision, Recall, F1-score, ROC-AUC, and PR-AUC. XGBoost and LightGBM followed closely, producing comparable predictive capability while requiring significantly shorter training times. Deep learning models such as TCN, LSTM, and GRU also demonstrated competitive performance; however, their computational cost was substantially higher than that of ensemble learning methods, making them less practical for large-scale deployment.

The dominance of ensemble algorithms confirms that combining multiple decision trees provides a more robust representation of the complex interactions present within the integrated urban crime dataset.

3.3 Comparison of Classification Metrics

To further evaluate the behaviour of individual models, the four primary classification metrics—Accuracy, Precision, Recall, and F1-score—were compared across all evaluated algorithms. These comparisons provide additional insight into the trade-offs between overall correctness, minority-class identification, and balanced predictive performance.

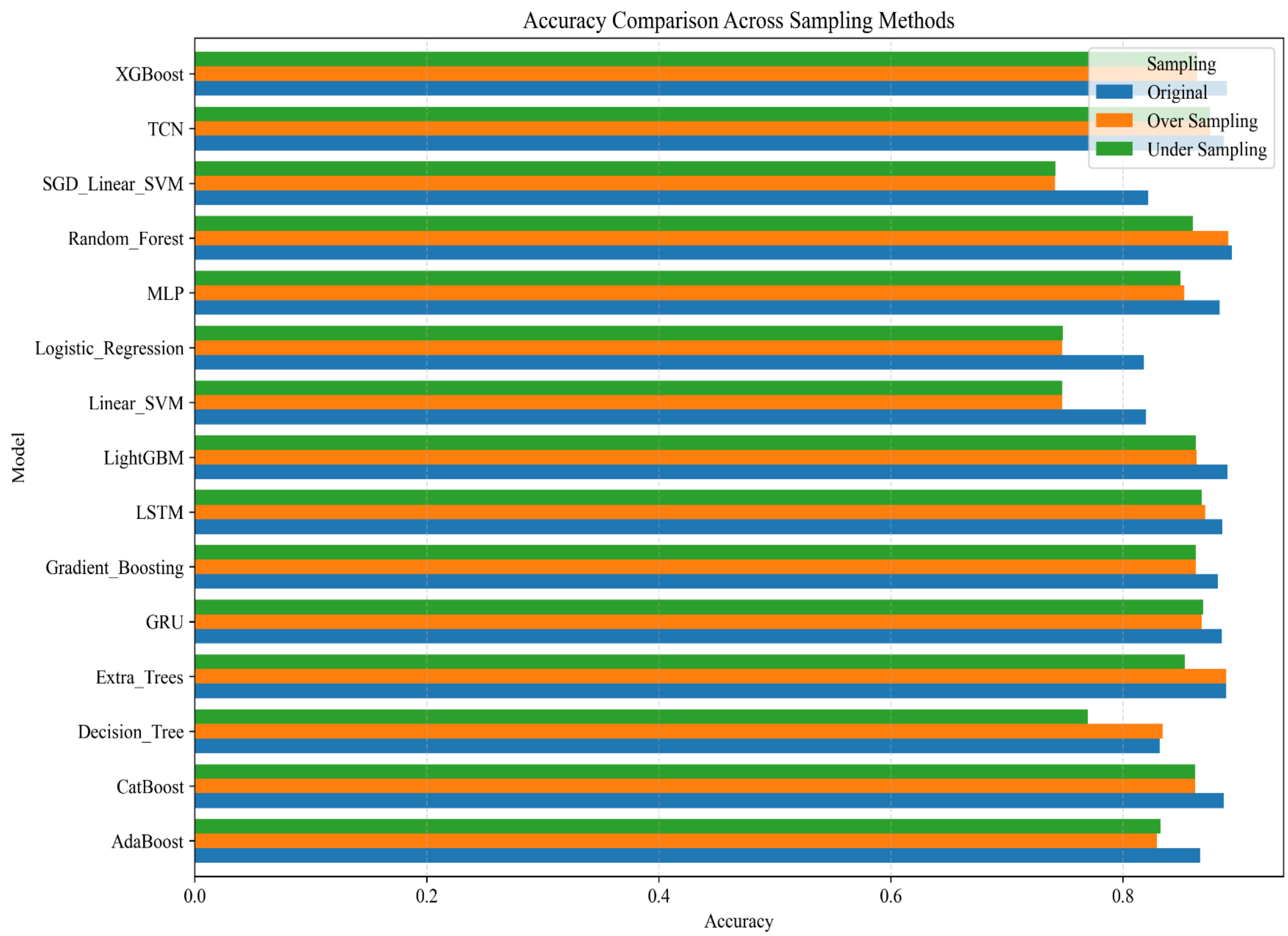

Figure 13: Accuracy Comparison Across Sampling Methods

The Accuracy comparison shows that several ensemble models achieved values exceeding 86%, with Random Forest producing one of the highest overall accuracies. Although Accuracy provides a useful measure of overall classification performance, it is strongly influenced by the majority non-arrest class. Consequently, high Accuracy alone should not be considered sufficient for evaluating arrest prediction models because it may conceal poor performance in identifying minority arrest cases.

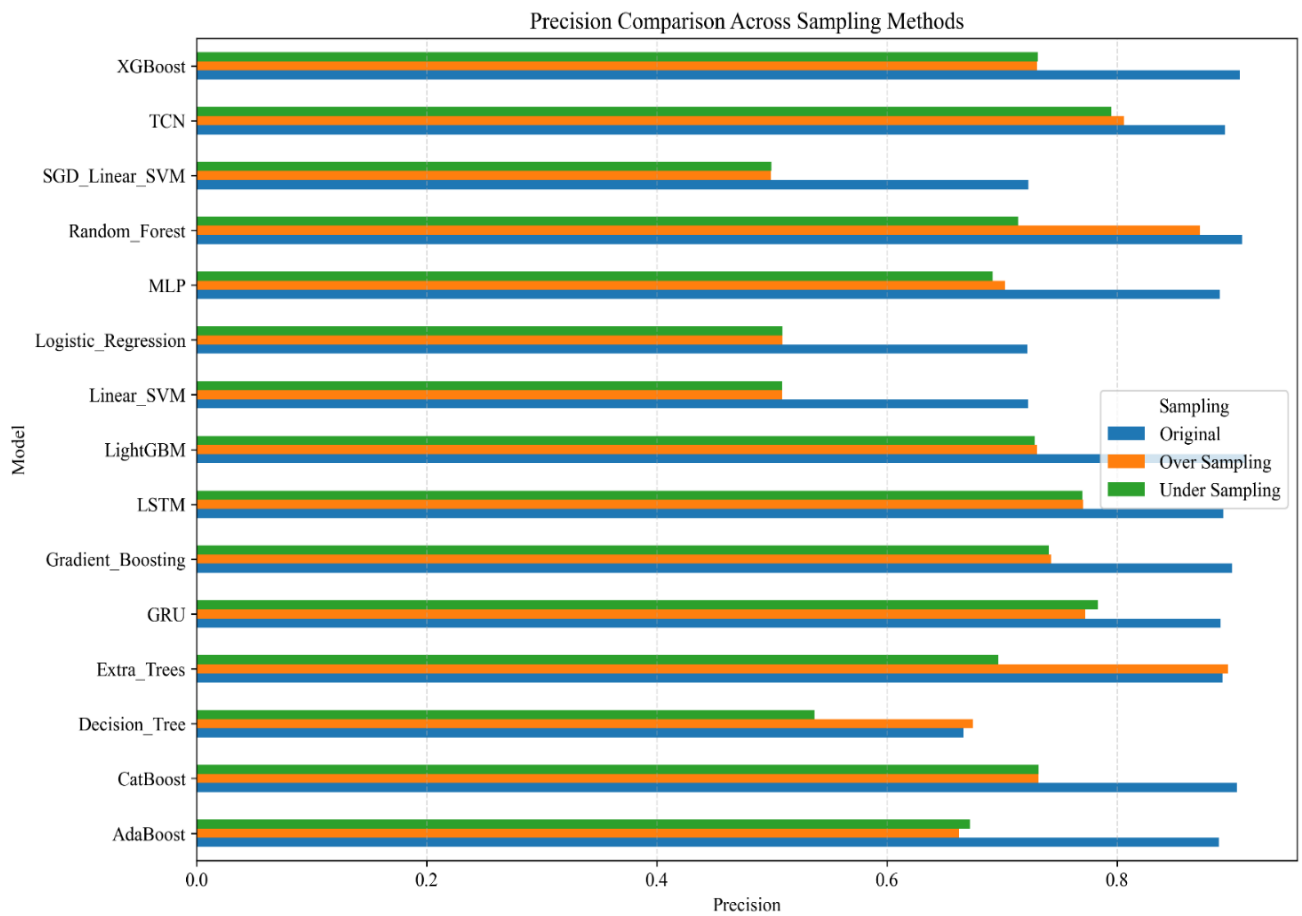

Figure 14: Precision Comparison Across Sampling Methods

Precision evaluates the proportion of predicted arrest cases that were correctly classified. Random Forest consistently achieved the highest Precision among the evaluated models, indicating that it generated relatively few false-positive predictions. XGBoost, LightGBM, CatBoost, and TCN also maintained high Precision values, demonstrating their ability to identify arrest cases with a high degree of confidence. This characteristic is particularly valuable for reducing unnecessary law enforcement interventions based on incorrect predictions.

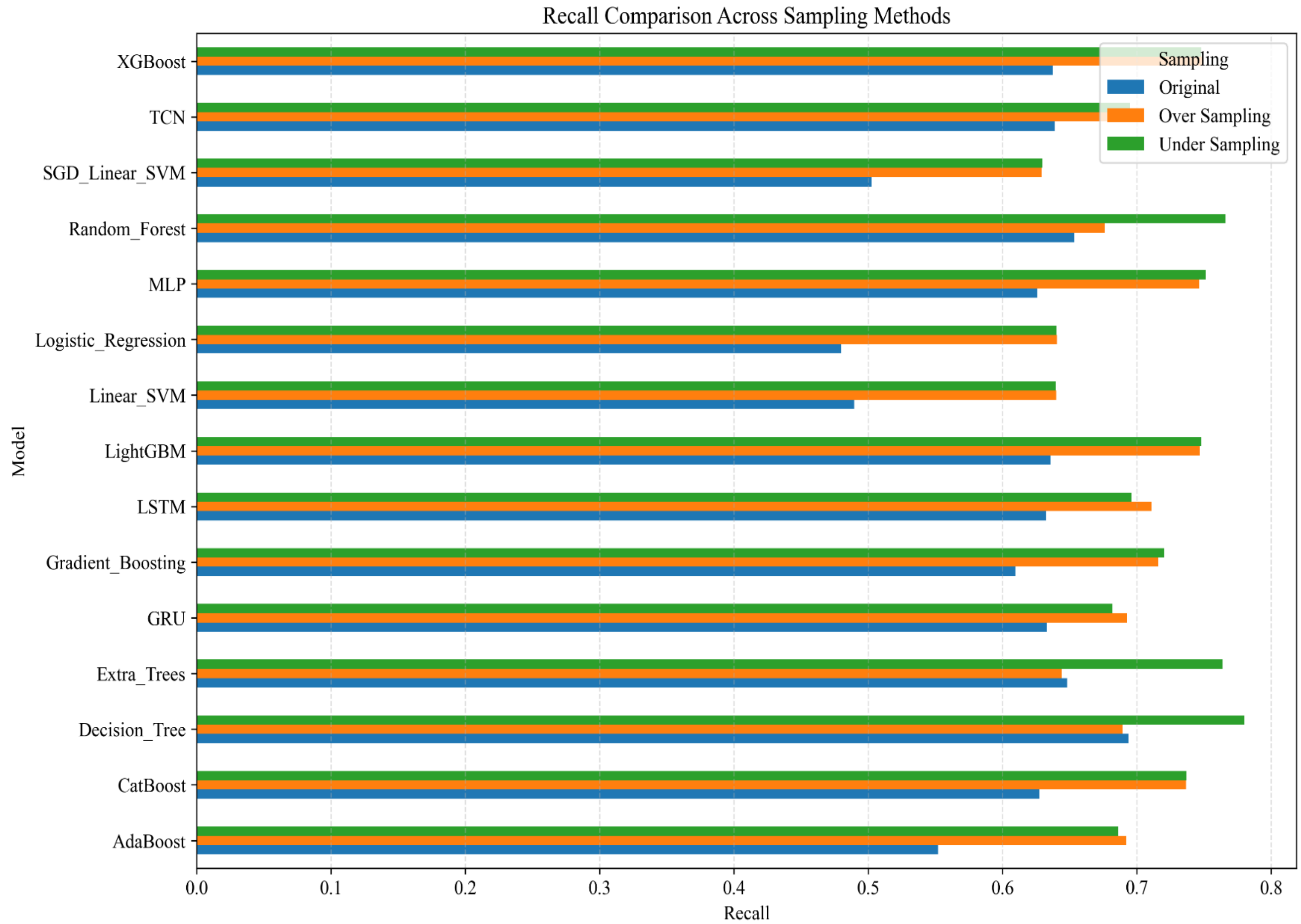

Figure 15: Recall Comparison Across Sampling Methods

Recall measures the proportion of actual arrest cases that were correctly identified by each classifier. As expected, both Random Undersampling and Random Oversampling substantially improved Recall compared with models trained on the original imbalanced dataset. Random Forest, XGBoost, LightGBM, Extra Trees, and MLP achieved the highest Recall values, demonstrating their ability to detect minority-class observations more effectively after class balancing.

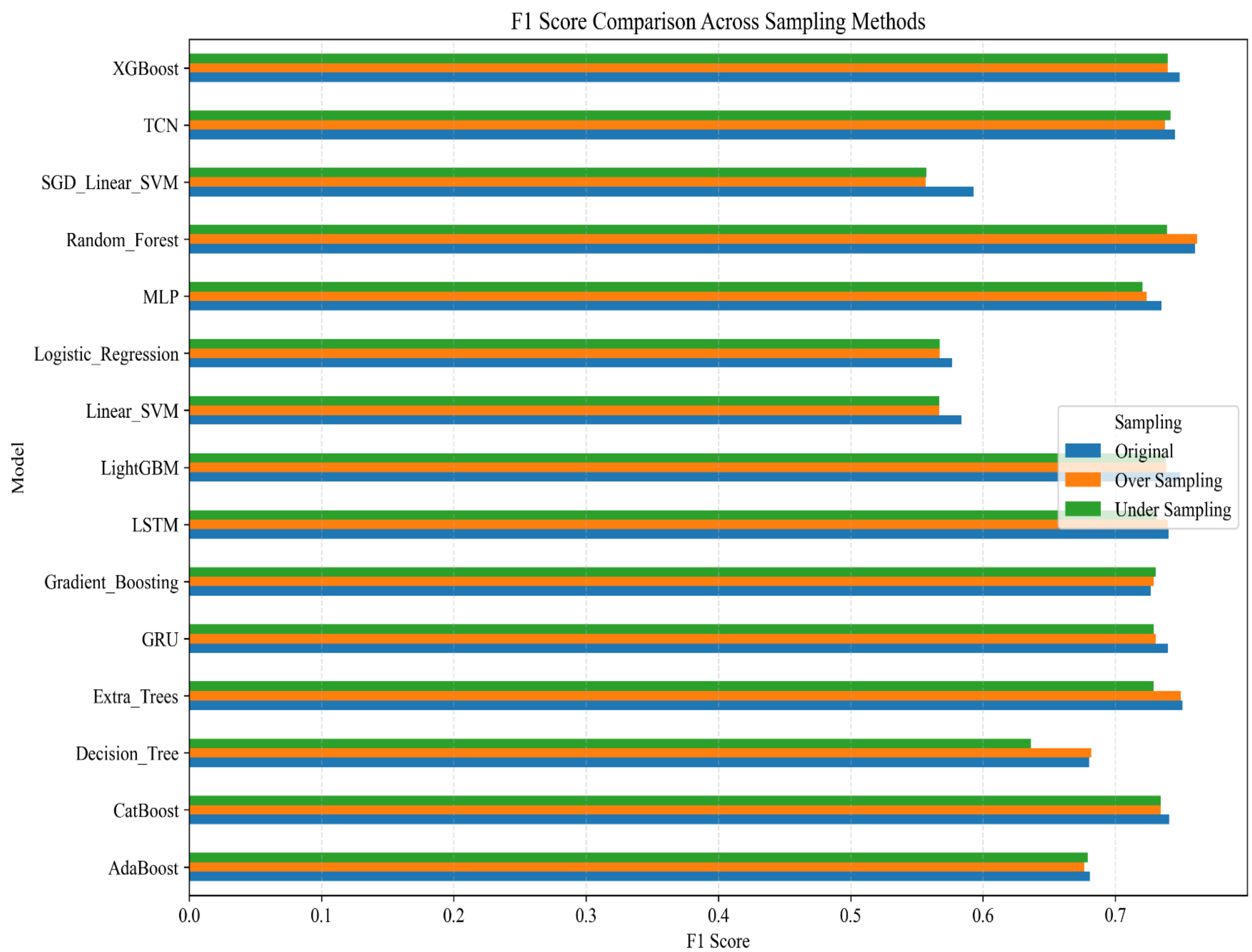

Figure 16: F1 Score Comparison Across Sampling Methods

Since arrest prediction represents an imbalanced classification problem, the F1-score provides the most appropriate overall evaluation by balancing Precision and Recall. Figure 16 shows that ensemble learning algorithms consistently achieved the highest F1-scores across all sampling strategies. Random Forest ranked first, followed closely by XGBoost, LightGBM, CatBoost, and TCN. The consistently strong F1-scores obtained by these models indicate that the feature set effectively captures the characteristics associated with arrest outcomes while maintaining a suitable balance between false-positive and false-negative predictions.

4 ROC-AUC and Precision–Recall Analysis

Although Accuracy, Precision, Recall, and F1-score provide valuable information regarding classifier performance, these metrics evaluate the models at a single classification threshold. In contrast, the Receiver Operating Characteristic Area Under the Curve (ROC-AUC) and Precision–Recall Area Under the Curve (PR-AUC) provide threshold-independent performance measures by evaluating model performance across all possible classification thresholds.

Since arrest prediction is an imbalanced binary classification problem, both ROC-AUC and PR-AUC were included to assess the overall discriminative ability of the developed models. ROC-AUC measures how effectively a classifier distinguishes between arrest and non-arrest cases, whereas PR-AUC evaluates the model's ability to correctly identify the minority arrest class while balancing precision and recall. Therefore, these metrics provide a more comprehensive comparison of model performance.

4.1 ROC-AUC Comparison

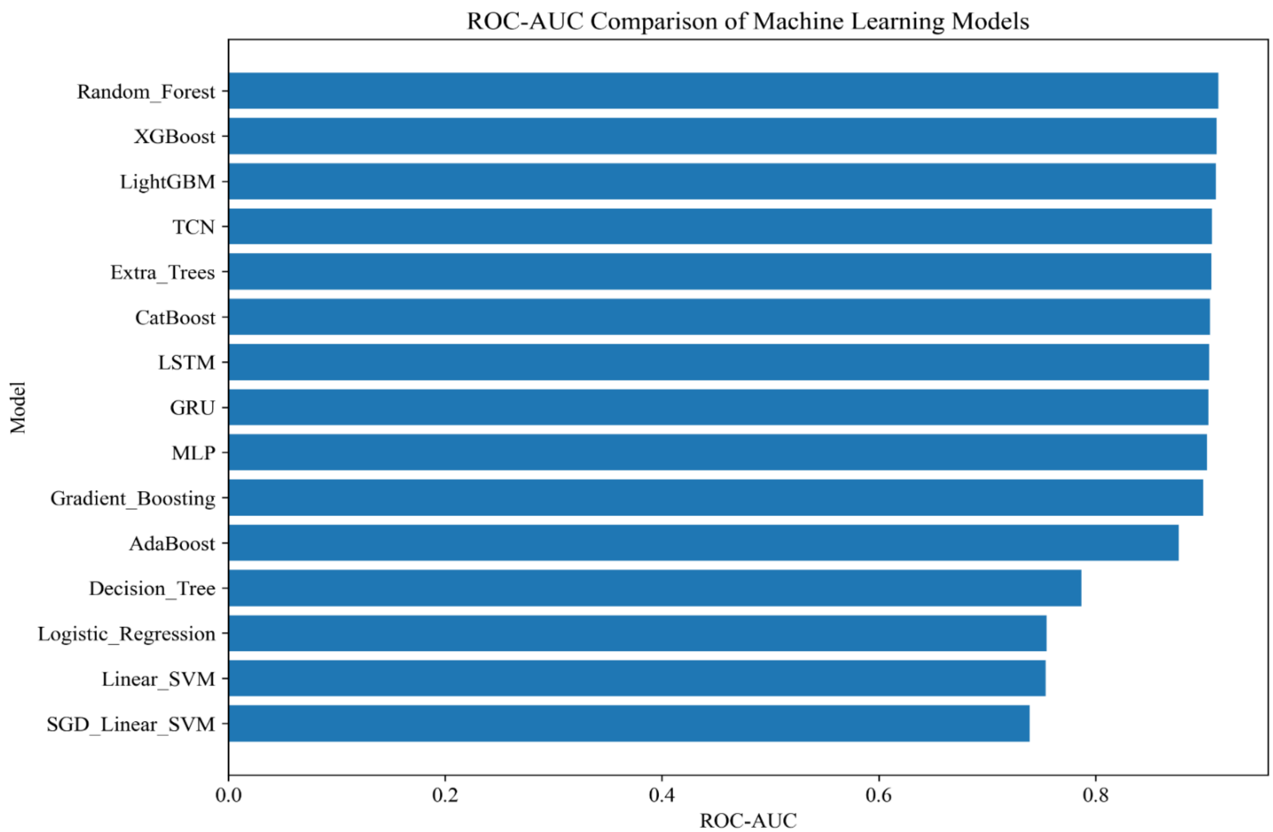

Figure 17: ROC Curves of the Best-Performing Models

Figure 17 compares the ROC-AUC values obtained by all evaluated machine learning and deep learning models. ROC-AUC measures the ability of a classifier to distinguish between arrest and non-arrest cases regardless of the selected decision threshold. A higher ROC-AUC value indicates stronger discriminative capability and more reliable classification performance.

The comparison shows that the ensemble learning algorithms consistently achieved the highest ROC-AUC values among all evaluated models. Random Forest obtained the highest ROC-AUC score, followed closely by XGBoost, LightGBM, CatBoost, Extra Trees, and MLP. All of these models achieved ROC-AUC values above 0.90, indicating excellent classification performance.

In contrast, traditional machine learning models such as Logistic Regression and Decision Tree produced comparatively lower ROC-AUC values, suggesting reduced capability in distinguishing arrest and non-arrest incidents.

The superior ROC-AUC performance of the ensemble models demonstrates their ability to effectively capture the complex relationships among crime characteristics, temporal variables, spatial information, socio-economic indicators, weather conditions, and public transit mobility features. These findings indicate that integrating multiple contextual features significantly improves the discriminatory power of arrest prediction models.

4.2 Precision–Recall (PR-AUC) Comparison

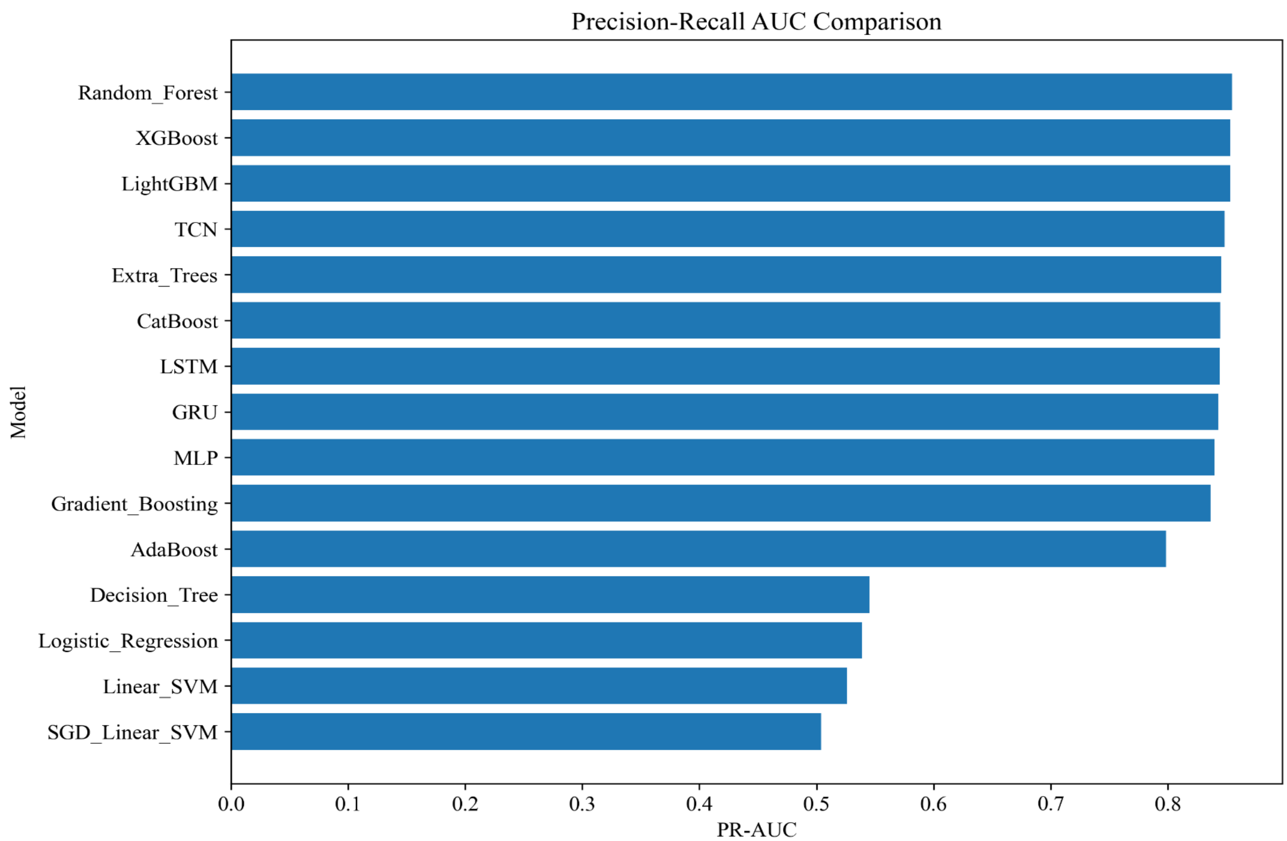

Figure 18: Precision–Recall Curves of the Best-Performing Models

Figure 18 presents the comparison of PR-AUC values for all evaluated classification models. Unlike ROC-AUC, PR-AUC focuses specifically on the prediction performance of the minority class and is therefore considered a more informative evaluation metric for imbalanced datasets. Since arrest cases constitute only approximately 25.8% of the observations, PR-AUC provides a more realistic assessment of model effectiveness in identifying arrest incidents.

The results indicate that Random Forest achieved the highest PR-AUC value, closely followed by XGBoost, LightGBM, CatBoost, Extra Trees, and MLP. Most of these ensemble models obtained PR-AUC values exceeding 0.85, demonstrating their ability to accurately identify arrest cases while maintaining a favorable balance between precision and recall. Conversely, Logistic Regression, Decision Tree, and several other baseline models produced relatively lower PR-AUC values, reflecting reduced effectiveness in detecting minority-class arrest cases.

Overall, the PR-AUC comparison further confirms the superiority of ensemble learning methods for arrest prediction. Their consistently high PR-AUC values indicate strong minority-class detection capability and lower susceptibility to the effects of class imbalance. Combined with the ROC-AUC comparison, these results demonstrate that ensemble learning algorithms provide the most reliable and robust predictive performance for the arrest prediction framework.

5 Discussion

The results demonstrate that the multi-source arrest prediction framework effectively models arrest outcomes by integrating crime-specific, temporal, spatial, socio-economic, weather, and public transit mobility features within a unified predictive framework. The exploratory analysis confirmed that each category of variables contributed complementary information, while the machine learning experiments showed that combining these heterogeneous data sources significantly improved predictive performance. The consistently strong results achieved by several ensemble learning models indicate that arrest prediction is influenced by complex interactions among multiple urban factors rather than by crime-related attributes alone.

One of the principal challenges encountered in this study was the naturally imbalanced distribution of arrest and non-arrest cases. The experiments demonstrated that relying solely on the original dataset may bias classifiers toward the majority class, despite producing high overall accuracy. The application of Random Undersampling and Random Oversampling improved minority-class recognition by increasing Recall and F1-score, confirming that addressing class imbalance is essential for developing reliable arrest prediction models. Although Random Oversampling produced the best overall predictive performance, Random Undersampling also provided competitive results with lower computational requirements.

Ensemble learning algorithms consistently outperformed traditional machine learning and deep learning approaches because they effectively capture complex, non-linear relationships among heterogeneous variables. Random Forest, XGBoost, LightGBM, Extra Trees, and CatBoost maintained an excellent balance between Precision and Recall while producing high ROC-AUC and PR-AUC values. In contrast, linear models such as Logistic Regression and Linear SVM were less capable of modelling the intricate interactions present in the integrated dataset. Although deep learning models, particularly the Temporal Convolutional Network (TCN), achieved competitive performance, the improvements were relatively small compared with the additional computational cost required for training.

From a practical perspective, the experimental results highlight the potential of data-driven arrest prediction for supporting evidence-based policing and strategic resource allocation. Accurate prediction of arrest outcomes can assist law enforcement agencies in identifying incidents that are more likely to result in an arrest, thereby enabling more efficient deployment of personnel and investigative resources. Moreover, incorporating contextual variables such as weather conditions, socio-economic characteristics, and public transit mobility enhances the models' ability to capture broader urban patterns associated with policing outcomes, contributing to more reliable and informed decision-making.

Despite the encouraging results, several limitations should be acknowledged. The study was conducted using crime data from a single metropolitan area, and therefore the generalizability of the framework to other cities requires further investigation. In addition, the models relied on historical records and contextual variables that may not capture all factors influencing arrest outcomes, such as officer deployment strategies or real-time operational information. Future research may explore advanced spatial–temporal deep learning architectures, graph neural networks, explainable artificial intelligence techniques, and real-time data integration to further improve predictive performance and interpretability.